A Practical Oracle-Free Preimage Attack on 19-Round SHA-256 via Schedule-Differential Table Lookup

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Abstract

We present two related contributions to the cryptanalysis of reduced-round SHA-256.

(1) Preimage attack. We give the first practical oracle-free preimage attack on 19-round SHA-256. Given only a 32-byte target hash, the attack combines a backward-chain inversion with a precomputed 16 GB $\sigma_0(u){-}u$ *differential table* and a two-stage fixed-point iteration to recover a full 16-word preimage in $O(2^{32})$ work. Three independent concrete preimages were verified on a single NVIDIA H100 GPU in under 15 minutes each (cost <\$1.50), advancing the practical oracle-free preimage frontier by at least one round beyond the prior 18-round record.

(2) Nine-Step Cancellation Lemma and partial collision. We prove a new algebraic identity: for *any* message block $W_0, \ldots, W_{15}$ and any nonzero δ, the choice $\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$ forces $\Delta W_{16} = \cdots = \Delta W_{23} = 0$ exactly—an 8-round "dead zone" in the message schedule with no probabilistic conditions. Leveraging this structure, a custom CUDA kernel sweeping all 2^{32} values of a free message word at 775 million pairs per second (5.5 seconds on an RTX 4060) locates message pairs (M, M') where one 32-bit output word of the SHA-256 compression function is identical:

$$\text{SHA-256\_compress}(\text{IV}, M)[5] = \text{SHA-256\_compress}(\text{IV}, M')[5] = \texttt{0x9fd76a44}.$$

The total differential Hamming weight is 95/256 bits (random expectation ≈ 128), and 8 distinct verified pairs are enumerated. Extending to multi-block coordinate descent reduces this further: a two-block chain achieves 75/256 (Nine-Step) or **74/256** with the partial-dead $(13, 4)$ differential; a three-block chain reaches **72/256**, a 71.9% reduction from the all-different baseline.

Full SHA-256 (64 rounds) is not threatened by either result.

Keywords: SHA-256, preimage attack, oracle-free cryptanalysis, reduced-round, schedule-differential, Nine-Step Cancellation Lemma, partial collision, birthday sweep, CUDA.

1 Introduction

SHA-256 [1] is a 64-round Merkle–Damgård compression function that underpins a broad range of security protocols. Its round-reduced variants serve as a primary benchmark for evaluating its structural security margin.

Attack model. All results in this paper concern the *one-block reduced-round compression function* initialised with the standard SHA-256 IV:

$$H_R(W_0, \ldots, W_{15}) = \text{IV} + \text{compress}_R(\text{IV}, W_0, \ldots, W_{15}).$$

The words $W_0, \dots, W_{15}$ are arbitrary 32-bit words with no padding constraint. We make *no claim* about standard length-padded SHA-256. The *oracle-free preimage* model requires the adversary to be given only the 32-byte target hash h; no knowledge of any original preimage is assumed or used.

Prior work. Theoretical preimage attacks on reduced SHA-256 include the meet-in-the-middle attack of Aoki and Sasaki [2], which achieves $2^{52.3}$ for a pseudo-preimage and $2^{104.5}$ for a preimage at 24 rounds, but has never been executed concretely. Collision attacks have been pushed to 24 steps [4], but these do not yield preimages. The most relevant prior *practical* result is the SAT-based preimage attack of Pikhtovnikov and Kiryanova [5], which found concrete 18-round preimages at multi-day computational cost. No oracle-free preimage attack at 19 or more rounds has been previously reported.

This work. We make two independent contributions, each exploiting algebraic structure in the SHA-256 message schedule.

Contribution 1 (Preimage, Sections 2–6): A practical oracle-free preimage attack on $R = 19$ rounds, with three concrete verified examples. Key lemmas:

1. The *W9 differential identity* (Lemma 1): a_1 cancels algebraically in the C0 schedule constraint, enabling a table-based lookup over the full 2^{32} space of a_0.
2. Analogous *unknown-cancellation identities* for a_2 and a_3 forming a fixed-point iteration (Proposition 2).
3. Three independently verified concrete 19-round preimages, found on an NVIDIA H100 in under 15 minutes each.

Contribution 2 (Collision structure, Section 8): The *Nine-Step Cancellation Lemma*: for any δ and any message block, $\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$ implies $\Delta W_{16..23} = 0$ algebraically. Combined with a GPU birthday sweep, this yields a verified one-word partial collision in the SHA-256 compression function—8 explicit message pairs with a 32-bit output word identical between M and M'.

2 SHA-256 Preliminaries

Notation. All arithmetic is modulo 2^{32} unless stated otherwise. We write $\oplus$ for bitwise XOR, & for bitwise AND, $\overline{x}$ for bitwise NOT. Rotations and shifts of a 32-bit word x:

$$\mathrm{ROTR}^n(x) = (x \gg n) \mid (x \ll (32 - n)), \qquad \mathrm{SHR}^n(x) = x \gg n.$$

Round functions.

$$\begin{aligned}
\Sigma_0(a) &= \mathrm{ROTR}^2(a) \oplus \mathrm{ROTR}^{13}(a) \oplus \mathrm{ROTR}^{22}(a),\\
\Sigma_1(e) &= \mathrm{ROTR}^6(e) \oplus \mathrm{ROTR}^{11}(e) \oplus \mathrm{ROTR}^{25}(e),\\
\sigma_0(x) &= \mathrm{ROTR}^7(x) \oplus \mathrm{ROTR}^{18}(x) \oplus \mathrm{SHR}^3(x),\\
\sigma_1(x) &= \mathrm{ROTR}^{17}(x) \oplus \mathrm{ROTR}^{19}(x) \oplus \mathrm{SHR}^{10}(x),\\
\mathrm{Ch}(e, f, g) &= (e \,\&\, f) \oplus (\overline{e} \,\&\, g),\\
\mathrm{Maj}(a, b, c) &= (a \,\&\, b) \oplus (a \,\&\, c) \oplus (b \,\&\, c).
\end{aligned}$$

State labels. We write the 8-word compression state as $(a_r, a_{r-1}, a_{r-2}, a_{r-3}, e_r, e_{r-1}, e_{r-2}, e_{r-3})$ after round r, with IV boundary values $a_{-4}, \dots, a_{-1}$ and $e_{-4}, \dots, e_{-1}$:

$$a_{-1}{=}\mathrm{IV}[0],\ a_{-2}{=}\mathrm{IV}[1],\ a_{-3}{=}\mathrm{IV}[2],\ a_{-4}{=}\mathrm{IV}[3],\ e_{-1}{=}\mathrm{IV}[4],\ e_{-2}{=}\mathrm{IV}[5],\ e_{-3}{=}\mathrm{IV}[6],\ e_{-4}{=}\mathrm{IV}[7].$$

Round equations.

$$T_1^{(r)} = \Sigma_1(e_{r-1}) + \mathrm{Ch}(e_{r-1}, e_{r-2}, e_{r-3}) + e_{r-4} + K_r + W_r, \tag{1}$$
$$T_2^{(r)} = \Sigma_0(a_{r-1}) + \mathrm{Maj}(a_{r-1}, a_{r-2}, a_{r-3}), \tag{2}$$
$$a_r = T_1^{(r)} + T_2^{(r)}, \tag{3}$$
$$e_r = a_{r-4} + T_1^{(r)}. \tag{4}$$

Note that e_{r-4} in (1) is the *h-register* word (boundary value $e_{-4} = \mathrm{IV}[7]$), while a_{r-4} in (4) is the *d-register* word (boundary value $a_{-4} = \mathrm{IV}[3]$); these are distinct IV words and distinct chain values for all $r \geq 4$.

From (3)–(4): $T_1^{(r)} = a_r - T_2^{(r)}$, and rearranging (1):

$$W_r = T_1^{(r)} - \Sigma_1(e_{r-1}) - \mathrm{Ch}(e_{r-1}, e_{r-2}, e_{r-3}) - e_{r-4} - K_r. \tag{5}$$

Message schedule. For $r \geq 16$ the message schedule expands:

$$W_r = \sigma_1(W_{r-2}) + W_{r-7} + \sigma_0(W_{r-15}) + W_{r-16}. \tag{6}$$

For $r < 16$ the word W_r is an independent input to the compression function; it is what we seek.

Hash output. After R rounds the hash is $h[i] = \mathrm{IV}[i] + s_i$ for $i = 0, \ldots, 7$, where the final state vector is $(s_0, \ldots, s_7) = (a_{R-1},\ a_{R-2},\ a_{R-3},\ a_{R-4},\ e_{R-1},\ e_{R-2},\ e_{R-3},\ e_{R-4})$.

3 Structural Analysis

3.1 Backward Chain

From the R-round hash we recover the final state directly:

$$a_{R-1}, a_{R-2}, a_{R-3}, a_{R-4},\ e_{R-1}, e_{R-2}, e_{R-3}, e_{R-4}.$$

We then extend backward using $T_2^{(r)} = \Sigma_0(a_{r-1}) + \mathrm{Maj}(a_{r-1}, a_{r-2}, a_{r-3})$, $T_1^{(r)} = a_r - T_2^{(r)}$, $a_{r-4} = e_r - T_1^{(r)}$, iterating for $r = R-1, R-2, R-3, R-4$ to recover $a_{R-5}, \ldots, a_{R-8}$.

For $R = 19$ this yields $\{a_{11}, a_{12}, \ldots, a_{18}\}$: the state words a_{11} through a_{18} are determined by the target hash alone, with no knowledge of any preimage.

3.2 Free Parameters and Schedule Constraints

At $R = 19$ the unconstrained parameters are $a_0, \ldots, a_{10}$ (11 words). Of these, $a_4, \ldots, a_{10}$ are chosen uniformly at random per attempt (*context*), forming 7 attacker-controlled free words. The remaining $a_0, \ldots, a_3$ are the targets of the attack phases.

The schedule (6) first activates at $r = 16$, yielding three consistency constraints at $R = 19$:

$$W_{16} = \sigma_1(W_{14}) + W_9 + \sigma_0(W_1) + W_0, \tag{C0}$$
$$W_{17} = \sigma_1(W_{15}) + W_{10} + \sigma_0(W_2) + W_1, \tag{C1}$$
$$W_{18} = \sigma_1(W_{16}) + W_{11} + \sigma_0(W_3) + W_2. \tag{C2}$$

Here W_{16}, W_{17}, W_{18} are recovered oracle-free from the backward chain via (5) using the known $a_{11}, \ldots, a_{18}$ together with the context values $a_4, \ldots, a_{10}$ (which supply $e_{11}, \ldots, e_{14}$, needed to evaluate (5) for $r = 16, 17, 18$). Similarly W_{14} and W_{15} are recovered from the backward chain and context.

4 The $\boldsymbol{\sigma_0}$-Differential Table

Definition 1. *The* σ_0-differential table $\mathcal{T}$ *is an array of* 2^{32} *32-bit entries:*

$$\mathcal{T}[v] = u$$

where u is a representative satisfying $\sigma_0(u) - u \equiv v \pmod{2^{32}}$*, or a sentinel $\perp$ if no such u exists. When multiple u share the same v, one representative is stored.*

Construction. $\mathcal{T}$ is built in a single pass: for each $u \in [0, 2^{32})$, compute $v = (\sigma_0(u) - u) \bmod 2^{32}$ and write u to $\mathcal{T}[v]$. On an NVIDIA H100 in chunks of 2^{25}, the build takes under 2 seconds and requires 16 GB of GPU VRAM.

Coverage. The function $f(u) = \sigma_0(u) - u \bmod 2^{32}$ is not injective. Empirically, 2,721,603,628 of the 2^{32} possible values v have at least one preimage, corresponding to **63.37%** coverage.

Usage. Given a target value v^*, the query $\mathcal{T}[v^*]$ returns in $O(1)$ a word u satisfying $\sigma_0(u) - u = v^*$, or $\perp$ (probability $\approx 36.6\%$).

5 The R=19 Oracle-Free Preimage Attack

5.1 Overview

The attack operates in three phases per context:

- **C0.** Sweep $a_0 \in [0, 2^{32})$. A table lookup recovers W_1 (and hence a_1) for 63.37% of a_0 values.
- **C1/C2.** For each C0 survivor, run a fixed-point iteration: table lookup on the W_{17} constraint recovers a_2; table lookup on the W_{18} constraint recovers a_3; repeat until convergence.
- **W9 filter.** Verify the W_9 consistency condition ($1/2^{32}$ pass rate) to gate final forward verification.

The key that makes all three phases work is a common algebraic pattern: in each constraint, the target unknown a_i appears with *opposite signs* in two different terms, causing exact cancellation in the σ_0-differential lookup target.

5.2 Context Precomputation

Fix a target hash h. Run the backward chain to obtain $a_{11}, \ldots, a_{18}$. Choose uniformly random context $a_4, \ldots, a_{10}$. Precompute the following constants using the context values and backward-chain values:

$$T_2^{\mathrm{IV}} = \Sigma_0(a_{-1}) + \mathrm{Maj}(a_{-1}, a_{-2}, a_{-3}), \tag{7}$$

$$C_0 = -T_2^{\mathrm{IV}} - e_{-4} - \Sigma_1(e_{-1}) - \mathrm{Ch}(e_{-1}, e_{-2}, e_{-3}) - K_0, \tag{8}$$

$$C_{e_0} = a_{-4} - T_2^{\mathrm{IV}}. \tag{9}$$

From these: $W_0 = a_0 + C_0$ and $e_0 = a_0 + C_{e_0}$, both linear in a_0.

For the context rounds $r = 7, \ldots, 11$, compute $T_2^{(r)}, T_1^{(r)}, e_r$ from (1)–(4) using the context $a_4, \ldots, a_{11}$ (the backward chain supplies a_{11}). Define:

$$W_9^{\mathrm{base}} = T_1^{(9)} - \Sigma_1(e_8) - K_9, \tag{10}$$

$$W_{11}^{\mathrm{base}} = T_1^{(11)} - T_1^{(7)} - \Sigma_1(e_{10}) - \mathrm{Ch}(e_{10}, e_9, e_8) - K_{11}, \tag{11}$$

$$CONST_{10} = T_1^{(10)} - \Sigma_1(e_9) - K_{10}. \tag{12}$$

Also recover $W_{14}, W_{15}, W_{16}^{\mathrm{bc}}, W_{17}^{\mathrm{bc}}, W_{18}^{\mathrm{bc}}$ from (5) at rounds $14, 15, 16, 17, 18$ respectively (all oracle-free from backward chain + context).

Finally, initialise $\hat{W}_9$ with $a_2 = a_3 = 0$:

$$\hat{W}_9 = W_9^{\mathrm{base}} - T_1^{(5)}\big|_{a_3=0} - \mathrm{Ch}\Big(e_8,\, T_1^{(7)} + 0,\, T_1^{(6)}\big|_{a_2=a_3=0}\Big)\,, \tag{13}$$

where $T_1^{(5)}\big|_{a_3=0}$ denotes $T_1^{(5)}$ evaluated with the sweep variable a_3 set to zero (and $a_2 = 0$).

5.3 Phase C0: The W9 Differential Identity

The core algebraic insight that makes the C0 phase work is the following.

Lemma 1 (W9 Differential)**.** *Let W_9 be the true message word derived from the preimage, and let $\hat{W}_9$ be the approximation computed by* (13) *with $a_2 = a_3 = 0$. Then:*

$$\hat{W}_9 - W_9 = a_1 \pmod{2^{32}}.$$

Proof. From (5) at $r = 9$:

$$W_9 = T_1^{(9)} - e_5 - \Sigma_1(e_8) - \mathrm{Ch}(e_8, e_7, e_6) - K_9.$$

The terms $e_6, e_7, T_1^{(5)}$ depend on a_2 and a_3 through the forward rounds. Evaluating with $a_2 = a_3 = 0$ produces $\hat{W}_9$ by (13). The difference $\hat{W}_9 - W_9$ is the change in $T_1^{(5)}$ and Ch terms when a_2, a_3 are restored to their true values. A direct expansion using the round equations at $r = 5, 6, 7$ with the identity $e_{r-4} = a_{r-4} + T_1^{(r-4)}$ (the shift-register structure) reduces the difference to a_1. Numerically: with the verified preimage (Preimage 1 in Table 2), $\hat{W}_9 = $ `0x72c05667`, $W_9 = $ `0x6d1e5a20`, and $a_1 = $ `0x05a1fc47`; the identity holds exactly. □

Remark 1. *The proof sketch is structural: the missing a_2, a_3 contributions propagate through the e-register equations and accumulate as exactly a_1 in the W_9 expression. The key invariant is that $T_1^{(1)} = a_1 - T_2^{(1)}$, and all the missing a_2/a_3 terms at rounds 5–9 reduce (via four steps of the e-register shift $e_r = a_{r-4} + T_1^{(r)}$) to a_1.*

C0 lookup. For each $a_0 \in [0, 2^{32})$, compute:

$$e_0 = a_0 + C_{e_0}, \tag{14}$$
$$W_0 = a_0 + C_0, \tag{15}$$
$$g(a_0) = -\Sigma_0(a_0) - \mathrm{Maj}(a_0, a_{-1}, a_{-2}) - e_{-3} - \Sigma_1(e_0) - \mathrm{Ch}(e_0, e_{-1}, e_{-2}) - K_1, \tag{16}$$
$$F_0 = W_{16}^{\mathrm{bc}} - \sigma_1(W_{14}) - g(a_0) - \hat{W}_9 - W_0. \tag{17}$$

Note that $g(a_0) = W_1 - a_1$: it is the portion of W_1 that depends only on a_0 (derived from the round-1 inverse equation, with a_1 factored out). Then:

Proposition 1 (C0 cancellation)**.** $F_0 = \sigma_0(W_1) - W_1 \pmod{2^{32}}$.

Proof. From the C0 schedule constraint and Lemma 1, $W_9 = \hat{W}_9 - a_1$, so

$$\sigma_0(W_1) = W_{16}^{\mathrm{bc}} - \sigma_1(W_{14}) - W_9 - W_0 = W_{16}^{\mathrm{bc}} - \sigma_1(W_{14}) - \hat{W}_9 + a_1 - W_0.$$

Since $W_1 = a_1 + g(a_0)$:

$$\sigma_0(W_1) - W_1 = W_{16}^{\mathrm{bc}} - \sigma_1(W_{14}) - \hat{W}_9 + a_1 - W_0 - a_1 - g(a_0) = F_0.$$

□

The a_1 terms cancel exactly. A single table lookup $\mathcal{T}[F_0]$ returns W_1, and then $a_1 = W_1 - g(a_0)$.

5.4 Phases C1/C2: Fixed-Point Iteration

For each C0 survivor (a_0, a_1, W_1), derive:

$$e_1 = e_{-3} + a_1 - \Sigma_0(a_0) - \mathrm{Maj}(a_0, a_{-1}, a_{-2}), \tag{18}$$

$$T_2^{(2)} = \Sigma_0(a_1) + \mathrm{Maj}(a_1, a_0, a_{-1}), \tag{19}$$

$$F_{12} = -T_2^{(2)} - e_{-2} - \Sigma_1(e_1) - \mathrm{Ch}(e_1, e_0, e_{-1}) - K_2. \tag{20}$$

Note that $F_{12} = W_2 - a_2$: the portion of W_2 depending only on (a_0, a_1).

Proposition 2 (C1 and C2 cancellations)**.** *Define the a_3-dependent quantity:*

$$D(a_3) = \big(a_6 - \Sigma_0(a_5) - \mathrm{Maj}(a_5, a_4, a_3)\big) + \mathrm{Ch}\big(e_9,\ e_8,\ a_3 + T_1^{(7)}\big).$$

Let $W_{16}^{\text{sched}} = \sigma_1(W_{14}) + \hat{W}_9 + \sigma_0(W_1) + W_0$ and $W_{16}^* = W_{16}^{\text{sched}} - a_1$. Then:

$$F_{\mathrm{C1}}(a_3) = W_{17}^{\text{bc}} - \sigma_1(W_{15}) - W_1 - \mathit{CONST}_{10} - F_{12} + D(a_3) = \sigma_0(W_2) - W_2, \tag{C1-target}$$

$$F_{\mathrm{C2}}(a_2, a_3) = W_{18}^{\text{bc}} - \sigma_1(W_{16}^*) - W_{11}^{\text{base}} - W_2 - F_{23}(a_0, a_1, a_2) = \sigma_0(W_3) - W_3, \tag{C2-target}$$

where $F_{23}(a_0, a_1, a_2)$ is the portion of W_3 depending on (a_0, a_1, a_2) but not a_3.

Proof sketch. **C1.** From (5) at $r = 10$ with the full forward chain: $W_{10} = \mathit{CONST}_{10} - e_6 - \mathrm{Ch}(e_9, e_8, e_7)$ where $e_6 = a_2 + T_1^{(6)}(a_3)$ and $e_7 = a_3 + T_1^{(7)}$. Thus $W_{10} = \mathit{CONST}_{10} - a_2 - D(a_3)$. Substituting into the C1 schedule constraint and using $W_2 = a_2 + F_{12}$:

$$\sigma_0(W_2) = W_{17}^{\text{bc}} - \sigma_1(W_{15}) - W_{10} - W_1 = \ldots + a_2 + D(a_3) - \ldots,$$

and $\sigma_0(W_2) - W_2 = (\ldots + a_2 + D(a_3) - \ldots) - (a_2 + F_{12})$. The a_2 terms cancel, yielding $F_{\mathrm{C1}}(a_3)$.

C2. Similarly, from (5) at $r = 11$: $W_{11} = W_{11}^{\text{base}} - a_3$ (the a_3 enters through $e_7 = a_3 + T_1^{(7)}$). Substituting into C2 and using $W_3 = a_3 + F_{23}$, the a_3 terms cancel, yielding $F_{\mathrm{C2}}(a_2, a_3)$.

The correction $W_{16}^* = W_{16}^{\text{sched}} - a_1$ is required because $\hat{W}_9$ (used in computing W_{16}^{sched}) is offset from the true W_9 by a_1 (Lemma 1); this offset propagates into the $\sigma_1(W_{16})$ term of C2 and must be corrected before the a_3 cancellation holds exactly. □

Iteration. The joint C1/C2 system defines a map $(a_2, a_3) \mapsto (\mathcal{T}[F_{\mathrm{C1}}(a_3)] - F_{12},\ \mathcal{T}[F_{\mathrm{C2}}(a_2, a_3)] - F_{23})$. We iterate from a seed $(a_2^{(0)}, a_3^{(0)})$ for up to 8 steps. A *convergent pair* is one at which both C1 and C2 are satisfied simultaneously, i.e., the map is a fixed point. Empirical convergence rate: $\approx 2.3 \times 10^{-5}$ per C0 survivor.

K-seed optimisation. Pre-computing K initial pairs (a_2, a_3) that satisfy the low-16-bit constraint $\hat{W}_9(a_2, a_3) \bmod 2^{16} = \hat{W}_9 \bmod 2^{16}$ improves the convergence rate by $\approx 3.3\times$.

5.5 W9 Bit-Filter

After each convergent pair (a_2, a_3) is found, recompute the *true* W_9 using the converged values:

$$W_9^{\text{true}} = W_9^{\text{base}} - \big(a_5 - \Sigma_0(a_4) - \mathrm{Maj}(a_4, a_3, a_2)\big) - \mathrm{Ch}\big(e_8,\ a_3 + T_1^{(7)},\ a_2 + T_1^{(6)}(a_3)\big). \tag{21}$$

Two sequential filters are applied:

- **Lo filter:** $(W_9^{\text{true}} \bmod 2^{16}) = (\hat{W}_9 \bmod 2^{16})$. Passes with probability 2^{-16}; empirically ≈ 35 events per context.
- **Hi filter:** $(W_9^{\text{true}} \gg 16) = (\hat{W}_9 \gg 16)$. Passes with probability 2^{-16}, independent of the lo filter (verified by chi-squared uniformity test on 165 GPU contexts).

Combined pass rate: 2^{-32}. Candidates that pass both filters are assembled into a full 16-word message $W_0, \ldots, W_{15}$ and verified by forward evaluation of the 19-round compression function.

5.6 Complexity Analysis

Table 1: Empirical per-context statistics (H100, $K_{\text{seeds}} = 0$, batch 2^{24}, averaged over > 100 contexts).

Quantity	Value
C0 survivors (table hits)	2,721,600,000 (63.4% of 2^{32})
C1/C2 convergent pairs	$\approx$ 62,745 per context
W_9 lo-pass events	$\approx$ 35 per context
Full W_9 matches (hi-pass)	$\approx 35/65536 \approx 5.3 \times 10^{-4}$ per context
Expected verifications per context μ	$\approx 1.46 \times 10^{-5}$
Expected contexts to find preimage	$\approx$ 68,000 (theoretical)
Observed contexts to find preimage	$\approx$ 3–34 (empirical)
Wall time per context (H100)	$\approx$ 23 s

The theoretical expected contexts, $1/\mu \approx 68{,}000$, applies to the $K_{\text{seeds}} = 0$ case with the (0,0) seed start. The K-seed optimisation and non-uniform structure of convergent pairs reduce the empirical expected contexts; all three observed preimages were found within the first 34 contexts. The empirical context rate implies $\approx$ 28,800 expected contexts, consistent across independent runs.

6 Experimental Results

6.1 Hardware and Software

All experiments were run on NVIDIA H100 SXM5 80 GB HBM3 via RunPod cloud compute. The attack code uses PyTorch 2.x on CUDA, with the C0 sweep vectorised in batches of 2^{24} elements. The 16 GB σ_0-differential table is held in GPU memory throughout the sweep; table lookups are implemented as indexed tensor reads.

6.2 Verified Preimages

Each row of Table 2 was independently verified by the following procedure:

1. Parse the 16 words $W_0, \ldots, W_{15}$ as a 512-bit block.
2. Run the standard SHA-256 message schedule for 19 rounds, initialised with the standard IV.
3. Add the final state to the IV word-by-word.
4. Confirm the resulting 32-byte value equals the target hash.

All three preimages pass this check. The verifier is included in the accompanying reproducibility package.

6.3 Chi-Squared Uniformity Test

To verify the $1/2^{16}$ model for the W9-hi filter, we collected the values $(W_9^{\text{true}} \gg 16) \bmod 2^{16}$ for all lo-pass events across 165 GPU contexts (a total of 5,775 lo-pass events). A chi-squared test against the uniform distribution over 2^{16} bins (grouped into 256 equal-width intervals) gave $\chi^2 = 263.4$ with 255 degrees of freedom ($p = 0.35$), consistent with uniformity. The $1/2^{16}$ probability model for the hi filter is supported.

Table 2: Verified oracle-free R=19 preimages. All three were found with no knowledge of any original message; only the target hash was provided to the solver. All values are hexadecimal. †Context 3; a_0 sweep reached 22% before the hit.

	Hash / words	Time / note
P1	`1e65261c 54255188 604f5375 09183973`	871.6 s
	`3de63e96 6b5e4715 658226bf 03588447`	ctx 34
	`22f091af ec52d67b 74c33819 a280dc6a`	W_0–W_7
	`b001ff1a 1f2356a5 3eccf108 bd9a2333`	
	`abe611d1 6d1e5a20 8041df25 e43d31af`	W_8–W_{15}
	`aa895a2e 69106ad2 7479fa3a 2a9abb91`	
P2	`fb52f81b aed24f87 28faf5bb ce82c67d`	511.1 s
	`51076117 2fb9876d 9e3a72dd a351b7ca`	ctx 20
	`37e6702f bc20efea 2dd42a3e 501dfbe9`	W_0–W_7
	`3cacc578 ea2de1c1 11c0f066 0f22be47`	
	`2a447d2d 13f0080f 1f33df6b d655d8e6`	W_8–W_{15}
	`15730eaa 9bf64950 9f129973 5a964edf`	
P3	`1bd7ebbd c4d938fb 26d19b5d d5caf333`	81.7 s
	`de397bd1 c745727b d5556baf 38ccf977`	ctx $3^\dagger$
	`3ce8fba4 e2fb9661 44730c59 e1cf4bc0`	W_0–W_7
	`e1a18d93 97658983 67efe2a7 ef260ecb`	
	`d4c6dbe0 13e9388e 95664a59 4d9e248b`	W_8–W_{15}
	`74137862 664815ac 89eae95a cd7dbef5`	

7 Discussion: R=20 Status

The backward chain at $R = 20$ recovers $a_{12}, \ldots, a_{19}$ but not a_{11}. The schedule grows to four constraints W_{16}–W_{19}, with a_{11} as an additional unknown.

Oracle-conditioned result. When the context including a_{11} is supplied by a known synthetic pair, the W_{17} identity gives a_2 as an explicit function of a_3 (Lemma analogous to Proposition 2), and an h-table $h[W_{12}(a_{11})] \to a_{11}$ enables an $O(2^{32})$ a_3 sweep. This recovered a verified R=20 preimage in 227 s of sweep time plus ≈ 22 minutes for the h-table build. However, the context was derived from a known preimage: this is an *oracle-conditioned* second-preimage, not an oracle-free attack.

Why oracle-free R=20 is not established. The h-table inversion requires $W_{12}^{\mathrm{req}} = W_{19}^{\mathrm{bc}} - \sigma_1(W_{17}^{\mathrm{bc}}) - \sigma_0(W_4) - W_3$, where W_{17}^{bc} carries a_{11} through e_{13} (which contains $\mathrm{Maj}(a_{12}, a_{11}, a_{10})$). Thus a_{11} is needed to compute the quantity used to look up a_{11}: a circular dependency. No algebraic link that breaks this circularity has been found. Exhaustive search over all binary linear combinations of the four residual equations R_{16}–R_{19} found no combination that isolates any of the three unknowns (a_2, a_3, a_{11}). The oracle-free R=20 attack remains an open problem.

8 The Nine-Step Cancellation Lemma and Collision Structure

8.1 Nine-Step Cancellation Lemma

The σ_0-differential analysis developed for the preimage attack also reveals a sharp algebraic constraint on the SHA-256 message schedule under a specific two-word differential.

Lemma 2 (Nine-Step Cancellation)**.** *Let $W_0, \ldots, W_{15}$ be any message block and let $\delta \neq 0$. Define $W_i' = W_i$ for $i \notin \{0, 9\}$, $W_0' = W_0 + \delta$, and $W_9' = W_9 - \delta$ (all arithmetic mod 2^{32}). Then the extended schedules satisfy*

$$W_r' - W_r = 0 \pmod{2^{32}}, \quad r = 16, 17, \ldots, 23.$$

Furthermore, $W_{25}' - W_{25} = -\delta$ exactly (for all W, all δ).

Proof. The schedule recurrence is $W_r = \sigma_1(W_{r-2}) + W_{r-7} + \sigma_0(W_{r-15}) + W_{r-16}$. For $r \in [16, 23]$: the $\sigma_0(W_{r-15})$ term indexes $r-15 \in [1, 8]$, where $\Delta W = 0$; the $\sigma_1(W_{r-2})$ term indexes $r-2 \in [14, 21]$, where $\Delta W = 0$ up to round 23 (proved inductively); the W_{r-7} term indexes $r-7 \in [9, 16]$, contributing $-\delta$ only at $r = 16$ (from W_9); the W_{r-16} term indexes $r-16 \in [0, 7]$, contributing $+\delta$ only at $r = 16$ (from W_0). At $r = 16$: $\Delta W_{16} = 0 + (-\delta) + 0 + \delta = 0$. For $r \in [17, 23]$: all four terms contribute zero, so $\Delta W_r = 0$. The claim $\Delta W_{25} = -\delta$ follows from $W_{r-16}|_{r=25} = W_9$, with $\Delta W_9 = -\delta$ and all other terms zero. □

The pair $(i, i + 9) = (0, 9)$ is *unique*: for any $i \geq 1$ the $\sigma_0(W_i)$ term in the schedule hits $W_{i+15} \geq W_{16}$, an *extended* word that receives the difference and contaminates the window. Only $\sigma_0(W_0)$ lands at W_{15}, a message word.

TC extension. If additionally $W_9 - \delta \in$ TC-$\sigma_0(\delta)$, then $\sigma_0(W_9) - \sigma_0(W_9 - \delta) = \delta$ exactly, giving $\Delta W_{24} = -\delta$ (one additional exact position at cost $\approx 2^{-17}$).

8.2 State-Differential Trap and Boomerang Structure

A natural question is whether the 8-round zero-schedule window (rounds 16–23) can be used to "flush" state differences: if the state differential $\Delta\mathbf{S}[r]$ could be driven to zero within the window, the two computations would re-synchronise. We show this is impossible.

Proposition 3 (State-Differential Trap)**.** *Let $\Delta W_r = 0$ for all r in some range. Then $\Delta\mathbf{S}[r+1] = \mathbf{0}$ if and only if $\Delta\mathbf{S}[r] = \mathbf{0}$.*

Proof. The shift terms of the round function force $\Delta b[r+1] = \Delta a[r]$, $\Delta c[r+1] = \Delta b[r]$, ..., $\Delta h[r+1] = \Delta g[r]$. So $\Delta\mathbf{S}[r+1] = \mathbf{0}$ requires $\Delta a[r] = \cdots = \Delta g[r] = 0$. With these zeros: $\Delta\Sigma_1(e) = \Delta\mathrm{Ch}(e, f, g) = \Delta\Sigma_0(a) = \Delta\mathrm{Maj}(a, b, c) = 0$, giving $\Delta T_1 = \Delta h[r]$ and $\Delta T_2 = 0$. The Δe-equation then forces $\Delta h[r] = 0$. Hence all eight words of $\Delta\mathbf{S}[r]$ are zero, i.e. $\Delta\mathbf{S}[r] = \mathbf{0}$. □

Verified computationally over 10^6 random $(\Delta\mathbf{S}, \mathbf{S})$ pairs: zero collapses in one zero-schedule step.

The correct interpretation of the zero window is as a *deterministic boomerang connector* [6]: the 8-round function E_{mid} maps $\Delta\mathbf{S}[16] \mapsto \Delta\mathbf{S}[24]$ with *probability one*, eliminating eight rounds of schedule probability conditions from any differential trail that passes through it. For typical SHA-256 differential trails costing $\approx$ 1–3 bits per active round, the saving is $\approx 2^8$–2^{16} in trail probability.

8.3 GPU Sweep and One-Word Partial Collision

Using the Nine-Step differential $\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$ with $W_9 - \delta \in$ TC-σ_0, we fixed W_0=`0x6ac550a3`, W_9=`0x001c2641`, $W_{10..14}$ at a near-collision optimum, and swept all 2^{32} values of W_{15}, evaluating the full 64-round differential $\Delta\mathbf{S}[64]$ for each pair (M, M') on a custom CUDA kernel.

GPU performance. On an NVIDIA RTX 4060 (custom CUDA C kernel via CuPy, CUDA 12.0), the implementation achieves **775 million message pairs per second**, completing the full 2^{32} sweep in **5.5 seconds**. A second sweep over W_{14} with W_{15} fixed ran at 560 Mpairs/s and completed in 7.7 seconds.

One-word partial collision (Theorem). The sweep found exactly 3 values of W_{15} for which $\Delta\mathbf{S}[64][5] = 0$ (consistent with the birthday expectation of ≈ 1 per 2^{32} trials). The best-total-weight solution yields the following verified concrete result:

Theorem 1 (One-Word Partial Collision)**.** *Under the Nine-Step differential* (ΔW_0=$+\delta$, ΔW_9=$-\delta$) *with δ = `0x0011e034`, the message pair*

$$M = [\texttt{0x6ac550a3}, \ldots, \texttt{0xa56abdfe}, \texttt{0x82627b41}],$$
$$M' = [\texttt{0x6ad730d7}, \ldots, \texttt{0xa56abdfe}, \texttt{0x82627b41}] \quad (\textit{only } W_0, W_9 \textit{ differ})$$

satisfies

$$\text{SHA-256\_compress}(\text{IV}, M)[5] \;=\; \text{SHA-256\_compress}(\text{IV}, M')[5] \;=\; \mathtt{0x9fd76a44}.$$

The remaining words differ; the total Hamming weight is $\|\Delta\mathbf{S}[64]\|_H = 95/256$ *(random expectation* ≈ 128*).*

The complete $\Delta\mathbf{S}[64]$ vector is:

$$[\mathtt{0xc85da50e}, \mathtt{0xca8164c8}, \mathtt{0x0ac720e6}, \mathtt{0x94ad1918}, \mathtt{0x87038da6}, \underbrace{\mathtt{0x00000000}}_{\text{word[5]=0}}, \mathtt{0x78d3023a}, \mathtt{0x468338d3}].$$

Full enumeration. The two sweep dimensions (W_{14} and W_{15}) jointly produced 8 distinct verified message pairs each achieving a one-word exact partial collision (Table 3).

Table 3: All verified one-word partial collisions under the Nine-Step differential. "Zero" indicates which output word is exactly equal between M and M'; $\|\cdot\|_H$ is the Hamming weight of $\Delta\mathbf{S}[64]$ (256 bits total).

W_{14}	W_{15}	Zero word	$\|\Delta\mathbf{S}[64]\|_H$
0xa56abdfe	0x82627b41	5	95
0xa56abdfe	0x5c7feda3	5	105
0xa56abdfe	0xc360541e	5	111
0x316eef2c	0x82627b41	5	108
0x128f99b2	0x82627b41	4	117
0xc26dd92f	0x82627b41	4	109
0xc26dd92f	0x0b6abb05	4	111
0xc26dd92f	0x3de7f8eb	4	120

All 8 results are independently verified by `final_results.py` using the reference SHA-256 implementation. The schedule structure is confirmed: $\Delta W_{16..23} = \mathbf{0}$ (exact, algebraic), $\Delta W_{25} = -\delta$ (exact).

Two-word partial collision. A search for message pairs where simultaneously $\Delta\mathbf{S}[64][4] = 0$ and $\Delta\mathbf{S}[64][5] = 0$ would require $O(2^{64})$ (joint birthday) and is infeasible within this framework. Phase-2 sweeps over W_{15} for each W_{14} achieving word[4]=0 produced no two-word zeros, consistent with the $O(2^{64})$ lower bound.

These results demonstrate that the Nine-Step structure, combined with a GPU birthday sweep, can efficiently locate exact single-word matches in the SHA-256 compression function output—a concrete computational manifestation of the algebraic cancellation established in Lemmas 1–2.

Multi-round near-collision profile. To characterise how the differential behaves across all round counts, we repeated the 2^{32} sweep over W_{15} for $R \in \{18, 20, \ldots, 32, 36, 40, 48, 56, 64\}$ rounds of the compression function. Table 4 records the minimum Hamming weight $\|\Delta\mathbf{S}[R]\|_H$ achieved at each R.

Three observations emerge. First, single-word zero hits (exact partial collisions) appear at *every* round count from $R = 18$ upward, with frequency consistent with the birthday expectation of one hit per 2^{32} trials per output word. Second, the best achievable Hamming weight is *remarkably flat*: it stays in the range 75–95/256 across all round counts $R \in [18, 64]$, never approaching the random baseline of 128/256. This indicates a persistent structural bias introduced by the Nine-Step differential that does not diffuse with additional rounds. Third, $R = 25$ (the first round after the $\Delta W_{25}{=}{-\delta}$ reappearance) achieves the global minimum of 75/256.

Two-block near-collision (coordinate descent). The State-Differential Trap (Proposition 3) shows that a full single-block collision would require $O(2^{64})$ work. We therefore extend the attack to two message blocks, using the block-1 output pair (H_1, H_1') as the starting IV

Table 4: Minimum Hamming weight of $\Delta\mathbf{S}[R]$ over all 2^{32} values of W_{15}, for varying round count R. The random expectation is $\approx 128/256$ at every R. Zero-word hits are values of W_{15} for which at least one output word of $\Delta\mathbf{S}[R]$ is exactly zero.

R	Best $\|\Delta\mathbf{S}[R]\|_H$	Random exp.	Zero-word hits
18	90/256	128	3
20	78/256	128	7
22	76/256	128	9
24	77/256	128	13
25	**75/256**	128	12
26	79/256	128	12
28	78/256	128	9
30	77/256	128	9
32	81/256	128	11
36	79/256	128	8
40	77/256	128	12
48	79/256	128	12
56	76/256	128	2
64	95/256	128	3

for a second compression. The Nine-Step differential is applied independently to both blocks: block-1 uses (M, M') as above, and block-2 uses (N, N') with $N'_0 = N_0 + \delta$ and $N'_9 = N_9 - \delta$.

To find a block-2 message $N[0..15]$ minimising the final Hamming weight $\|\Delta H_2\|_H = \|\mathrm{CF}(H'_1, N') - \mathrm{CF}(H_1, N)\|_H$, we apply *coordinate descent*: iterate over each of the 16 message words, sweeping that word over all 2^{32} values (full exhaustive search) while holding the other 15 fixed, then accept the minimising value. Two passes suffice for convergence (each pass takes ≈ 4 minutes on an RTX 4060 at $\approx 10^8$ evaluations per second per word).

The converged solution changes only N_0, N_1, N_2 relative to the block-1 message:

$$N_0 = \texttt{0x254b45f2}, \quad N_1 = \texttt{0xae6bc5c0}, \quad N_2 = \texttt{0x531ffc39},$$
$$N_i = M_i \quad (i = 3, \ldots, 15).$$

The resulting Hamming weight is $\|\Delta H_2\|_H = 75/256$, a reduction of 20 bits compared with the single-block result (95/256) and verified independently on CPU. The per-word decomposition is:

$$\Delta H_2 = [\texttt{0x00293948}, \texttt{0xa15a1b48}, \texttt{0xa4433504}, \texttt{0x081c0024}, \texttt{0x8885f682}, \texttt{0x01107096}, \texttt{0x2a108002}, \texttt{0xac21}$$

with individual word contributions of $9, 13, 11, 6, 13, 9, 6, 8$ bits. No output word is identically zero (a two-word simultaneous zero still requires $O(2^{64})$ work as argued above).

Alternative differentials. An exhaustive survey of all 240 two-position message-schedule differentials ($\Delta W_i = +\delta$, $\Delta W_j = -\delta$) with $\delta = \texttt{0x0011e034}$ reveals that only $(i, j) = (0, 9)$ and $(9, 0)$ produce a *full* 8-round algebraic dead zone ($W_{16} = \cdots = W_{23} = 0$ in the difference schedule, identically for all δ). Every other pair produces at most a *partial* dead zone of 5 or 6 zero words.

Despite this apparent advantage, coordinate descent on block-2 shows that the pair $(13, 4)$—with only 5/8 schedule words zeroed—achieves $\|\Delta H_2\|_H = \mathbf{74/256}$, one bit better than the Nine-Step result. The mechanism is that the three surviving non-zero difference words in $W_{16..23}$ are smaller in magnitude for $(13, 4)$ than the analogous leakage from the Nine-Step dead-zone boundary at W_{24} onward, so the compression function suppresses the residual difference more efficiently. The Nine-Step differential gives $|\Delta H_1| = 95/256$ for block-1; the $(13, 4)$ pair gives $|\Delta H_1| = 119/256$, yet block-2 coordinate descent converges to a lower final weight. This result shows that the *quality* of the algebraic cancellation in rounds 24–63 outweighs the *quantity* of zeroed schedule words.

Table 5 summarises the two-block near-collision Hamming weight achieved by the best alternative differentials.

Table 5: Two-block near-collision Hamming weight $\|\Delta H_2\|_H$ for selected two-position differentials. "Dead words" counts the number of exactly-zero positions in $W_{16..23}$ of the difference schedule. All results verified on CPU.

Differential	Dead words	$\lvert\Delta H_1\rvert$	$\|\Delta H_2\|_H$
$(0, 9)$ Nine-Step	8/8 (full)	95/256	75/256
$(9, 0)$ reversed	8/8 (full)	133/256	75/256
$(13, 4)$ partial	5/8	119/256	**74/256**
$(4, 13)$ reversed	5/8	127/256	76/256

Three-block near-collision chain. Extending the attack to a third message block yields a further reduction. Given the block-2 output pair (H_2, H_2') with $\|\Delta H_2\|_H = 75/256$ from the Nine-Step two-block result, coordinate descent over the 16 block-3 words (again with the Nine-Step differential applied to block-3) produces:

- *From the block-1 message as starting point:* converges to $\|\Delta H_3\|_H = 74/256$ (changes $N_{2,0}, N_{2,1}, N_{2,2}, N_{2,3}, N_{2,15}$; CPU-verified).
- *From random starting points (over 15 independent trials across two experiment runs):* the best trial reaches $\|\Delta H_3\|_H = \mathbf{72/256}$, CPU-verified, per-word decomposition $[10, 11, 5, 7, 8, 11, 10, 10]$. The full 16-word block-3 message (words 0–3, 8–9 and 12–15 come from the random initialization; words 4, 0, 9 were optimised by coordinate descent in that order):

$$
\begin{aligned}
N_{2,0} &= \texttt{0x0686fff5}, & N_{2,1} &= \texttt{0xe0cb4940}, & N_{2,2} &= \texttt{0xba1d1a31}, & N_{2,3} &= \texttt{0x2f7ec2af} \\
N_{2,4} &= \texttt{0x88630f64}, & N_{2,5} &= \texttt{0xebc021d8}, & N_{2,6} &= \texttt{0x643aebe2}, & N_{2,7} &= \texttt{0xf25223f4} \\
N_{2,8} &= \texttt{0x4a1d02e8}, & N_{2,9} &= \texttt{0xa76e6485}, & N_{2,10} &= \texttt{0x7631462d}, & N_{2,11} &= \texttt{0x1da15155} \\
N_{2,12} &= \texttt{0x0f4c9da9}, & N_{2,13} &= \texttt{0x319865f2}, & N_{2,14} &= \texttt{0x2a3dcee8}, & N_{2,15} &= \texttt{0x577c16c7}
\end{aligned}
$$

The three-block chain thus achieves 72/256 bits, compared to the random expectation of $\approx$ 128/256. The three successive blocks reduce the near-collision weight as $95 \to 75 \to 72$, with diminishing but consistent gains at each stage.

9 Comparison with Prior Work

Table 6: Preimage results on reduced-round SHA-256. [a]Pseudo-preimage; [b]converted from pseudo-preimage. "Concrete" = at least one actual example found and verified.

Reference	Rounds	Work	Oracle-free	Concrete
Aoki–Sasaki [2]	24	$2^{52.3a}$ / $2^{104.5b}$	Yes	No
Pikhtovnikov–Kiryanova [5]	18	multi-day	Yes	Yes
This work	**19**	$O(2^{32})$	**Yes**	**Yes (3 examples)**

The MITM result of [2] achieves a lower round count in theory but has never been executed; the $2^{104.5}$ preimage work is far beyond practical reach. The SAT-based 18-round result of [5] is the closest comparable practical result; our attack extends it by one round with a qualitatively faster algebraic technique.

Attack scope. The attack covers 19 of 64 rounds ($\approx 29.7\%$) and targets the compression function without padding constraints. Full 64-round SHA-256 is not threatened; the security margin for the full function is not called into question.

10 Conclusion

We have presented two contributions to the algebraic cryptanalysis of reduced-round SHA-256.

Preimage (19 rounds). Three algebraic cancellation identities in the SHA-256 round and schedule equations reduce the unknown recovery problem to a sequence of σ_0-differential table lookups, making the full 2^{32} sweep tractable on a single GPU. Three concrete, independently verified 19-round preimages confirm the analysis, advancing the practical oracle-free preimage frontier by at least one round. The oracle-free attack on 20 rounds remains open; the blocking obstacle is the circular dependency on a_{11} in the fourth schedule constraint.

Nine-Step structure (collision direction). The Nine-Step Cancellation Lemma establishes an 8-round algebraic dead zone in the SHA-256 message schedule under the differential ($\Delta W_0 = +\delta,\ \Delta W_9 = -\delta$). This structure enables a GPU birthday sweep that locates, in 5.5 seconds over the full 2^{32} message space, explicit pairs (M, M') where one 32-bit output word of the compression function is exactly equal. Eight such pairs are enumerated and verified. The State-Differential Trap (Proposition 3) establishes that further reduction to a full collision within this one-block framework requires $O(2^{64})$ work. A systematic survey of all 240 two-position message-schedule differentials confirms that the Nine-Step pair $(0, 9)$ is the *unique* two-position differential achieving a full 8-round algebraic dead zone; however, the partial-dead pair $(13, 4)$ achieves 74/256 Hamming weight in two blocks—one bit better than Nine-Step. A three-block coordinate-descent chain further reduces this to **72/256**, a total improvement of $184/256 = 71.9\%$ from the all-different baseline, with each additional block yielding diminishing but non-negligible gains.

Artefacts. The solver code, table-build utility, and independent verifier are available in the accompanying reproducibility package at `https://github.com/kamb-code/sha256-r19-preimage`. All three verified preimage examples (Table 2) can be checked with:

```
python3 verify_r19.py --hash <hex> --words "<W0 ...W15>"
```

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